CURRICULUM VITAE

Application for the Post of Research Fellow (Information Technology)

Personal Information

- **Name:** Swati Aggrawal
- **Domain Applied For:** Research Fellow (Information Technology)
- **Date of Birth:** 13/03/2001
- **Nationality:** Indian
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Academic Qualifications

S.No.	Degree	Year	Institution	Subjects/Coursework	CGPA	Distinction
1.	M.Sc. Computer Science	2026	Sapienza University of Rome	Dissertation: Evaluating Pretrained Vision Transformers Under Continual Learning	91/110	First Division
2.	B.Tech Computer Science Engineering (AI & ML)	2023	Sharda University	Computer Science Engineering with specialization in AI and ML	8.93/10	First Division
3.	Class 12	2019	Ram-Eesh International School	PCM	91.2/100	First Division
4.	Class 10	2017	Ram-Eesh International School	Mathematics, Science, SST, Hindi, English	10/10	First Division

Professional Experience

Researcher

Vision Lab, Sapienza University of Rome, Italy

Employment Type: Researcher (Master's Research)

Duration: November 2025 – May 2026 (7 months)

Supervisor: Prof. Marco Raoul Marini

Responsibilities

- Conducted research on continual learning for computer vision under the supervision of Prof. Marco Raoul Marini.
- Investigated catastrophic forgetting in sequential learning and evaluated parameter-efficient adaptation techniques for Vision Transformers.
- Implemented and evaluated adapter-based continual learning approaches integrated with Elastic Weight Consolidation (EWC).
- Performed experimental evaluation of pre-trained Vision Transformer and ResNet architectures on sequential image classification tasks.
- Compared model performance using continual learning metrics including Average Accuracy, Average Forgetting, Backward Transfer (BWT), and Forward Transfer (FWT).
- Conducted literature review and comparative analysis of regularization-based, replay-based, architectural, and hybrid continual learning methods.
- Prepared technical documentation, experimental reports, and a master's dissertation.

Programming Languages Used

- Python

Libraries & Frameworks

- PyTorch
- NumPy
- Matplotlib
- timm (PyTorch Image Models)

Development Environment & Tools

- Google Colab
- Git
- LaTeX

Models Evaluated

- Vision Transformer (ViT-Tiny)
- ResNet-18
- Adapter-based Vision Transformers
- Elastic Weight Consolidation (EWC)

Datasets Worked With

- Split CIFAR-100 (benchmark for continual learning experiments)

Evaluation Metrics

- Average Accuracy
- Average Forgetting
- Backward Transfer (BWT)
- Forward Transfer (FWT)

Research Outputs

- Master's Dissertation: *Evaluating Pretrained Vision Transformers Under Continual Learning*
- Experimental evaluation of continual learning methods for Vision Transformers
- Comparative performance analysis of CNN- and Transformer-based continual learning approaches

Projects

1. Metal Surface Defect Detection

Technologies: Python, PyTorch, YOLOv8, OpenCV

Project Link: *https://github.com/swatiaggrawal/Metal-Defect-Detection*

Description

- Developed a computer vision system for automated detection of surface defects in metal components using deep learning.
- Trained and evaluated a YOLOv8-based object detection model to identify and localize defects from industrial images.
- Performed image preprocessing, model training, validation, and performance evaluation.
- Utilized OpenCV for image processing and visualization of detection results.
- Designed the solution to support automated quality inspection in manufacturing environments.

2. Document Change Tracker

Technologies: Python, FastAPI, SentenceTransformers, Docker, AWS ECS/Fargate, CloudWatch

Project Link: *https://github.com/swatiaggrawal/DocDelta*

Description

- Developed an AI-powered document comparison system to identify and summarize changes between document versions.
- Implemented semantic similarity using SentenceTransformers to detect modifications beyond exact text matching.
- Built RESTful APIs using FastAPI for document processing and comparison.
- Containerized the application using Docker and deployed it on AWS ECS/Fargate with CloudWatch monitoring.
- Designed the system for scalable document analysis and automated change tracking.
- Designed and implemented efficient document indexing and retrieval workflows for processing large collections of textual data using semantic embeddings and REST APIs.

Publications

Aggrawal, S., & Bhushan, B. (2023)

Machine Learning for Air Pollution. *2023 2nd International Conference on Vision Towards Emerging Trends in Communication and Networking Technologies (ViTECoN), IEEE.*

DOI: 10.1109/ViTECoN58111.2023.10157028

Link: *https://doi.org/10.1109/ViTECoN58111.2023.10157028*

Description

- Conducted a comprehensive review of machine learning techniques for air quality evaluation and prediction.
- Examined six widely used machine learning models, including Random Forest, Support Vector Machine (SVM), Decision Tree, Artificial Neural Network (ANN), Genetic Algorithm (GA), and Least Squares SVM (LS-SVM).
- Reviewed prediction-based and knowledge discovery approaches used in air pollution epidemiology.
- Analyzed air quality assessment based on the six National Ambient Air Quality Standards (NAAQS) criteria pollutants: $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, and O_3.
- Discussed recent advances, limitations, and future research directions in applying machine learning to air quality monitoring and prediction.
- Published in the IEEE proceedings of the 2023 International Conference on Vision Towards Emerging Trends in Communication and Networking Technologies (ViTECoN).

Technical Skills

- **Programming Languages:** Python, Java, SQL, C++, Bash
- **Data Analysis & Scientific Computing:** Pandas, NumPy, Matplotlib, Statistical Analysis, Data Preprocessing, Experimental Design, Model Evaluation, Data Visualization
- **Machine Learning & Artificial Intelligence:** Machine Learning, Deep Learning, Computer Vision, Vision Transformers (ViTs), Continual Learning, Transfer Learning, Hugging Face Transformers, PyTorch, TensorFlow, Keras, Scikit-learn
- **Software Development & APIs:** FastAPI, REST API Development
- **Databases:** SQL, Database Querying and Management
- **Cloud & DevOps:** Docker, AWS ECS/Fargate, Amazon CloudWatch, Git, Linux
- **Research & Documentation:** Research Methodology, Literature Review, Technical Writing, Experimental Design, Performance Benchmarking, Scientific Documentation, Presentation of Research Findings

Experience Working with National/International Bodies

Sapienza University of Rome, Italy (International)

Role: Researcher, Vision Lab
Duration: November 2025 – May 2026

Worked as a Researcher in the Vision Lab under the supervision of **Prof. Marco Raoul Marini**, contributing to research on continual learning for computer vision. Responsibilities included implementing and evaluating machine learning models, conducting experimental analysis, reviewing scientific literature, and preparing technical documentation and a Master's dissertation.

IEEE (International Professional Organization)

Role: Author

Published the research paper **"Machine Learning for Air Pollution"** in the proceedings of the **2023 2nd International Conference on Vision Towards Emerging Trends in Communication and Networking Technologies (ViTECoN), IEEE**. The publication reviewed machine learning techniques for air quality prediction and evaluation, covering major algorithms, learning paradigms, air quality standards, and future research directions.

Brief Note on Suitability

I hold an M.Sc. in Computer Science from Sapienza University of Rome and a B.Tech in Computer Science Engineering (Artificial Intelligence and Machine Learning). My research experience includes developing and evaluating machine learning models using Python, conducting experimental analysis, and preparing technical documentation. Through my academic coursework and research, I have developed skills in programming, data analysis, and software development. I am interested in applying these skills to support the research and analytical work of the Tax Policy Research Unit.

Declaration

I hereby declare that the information furnished above is true and correct to the best of my knowledge and belief.

Date: 14/07/2026

Place: Greater Noida

Signature: SWATI AGGRAWAL